2.1.2.3.2 Remote Desktop and Web Services

In Windows, the Remote Desktop Protocol (RDP) [MS-RDPBCGR] and the Web Services Management Protocol [MS-WSMV] use the CredSSP Protocol to delegate the user's credentials from the client to the target server.

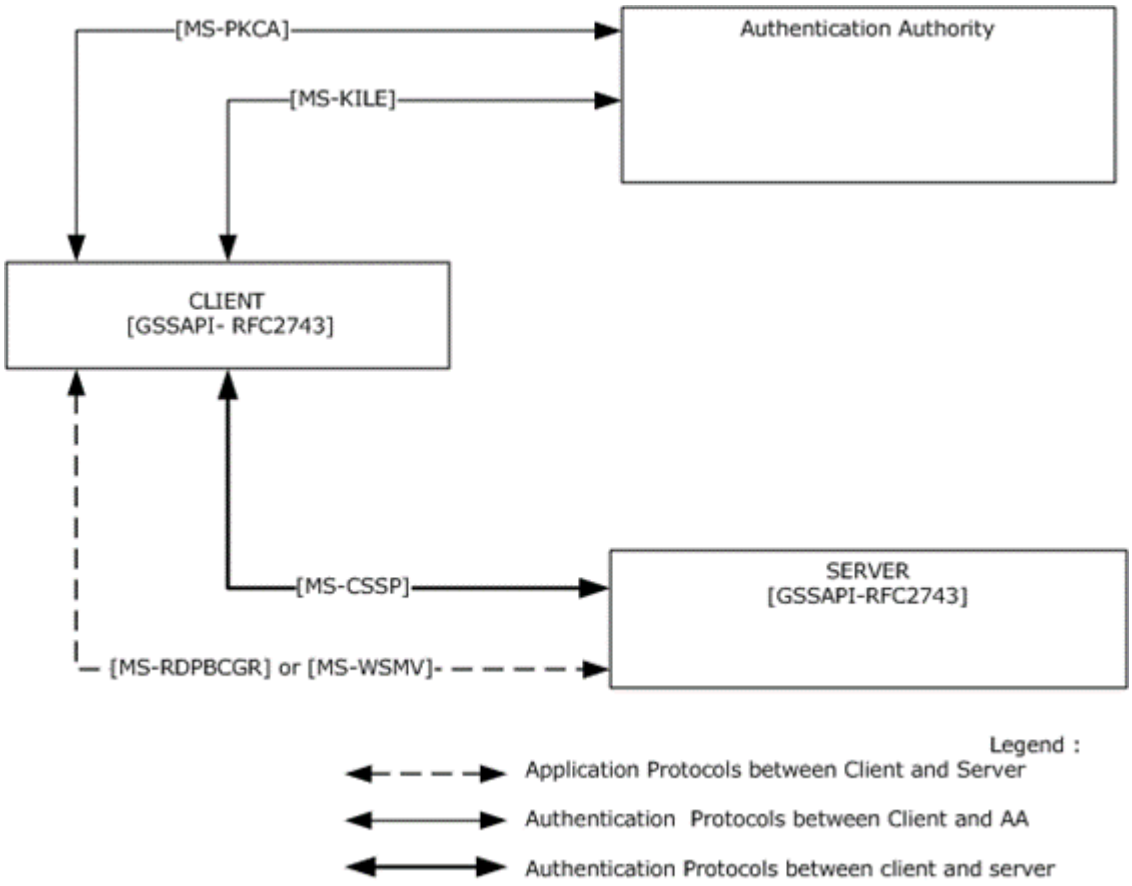


Figure 9: Credential delegation through the CredSSP Protocol

The preceding diagram shows that RDP and the Web Services Management Protocol trigger the CredSSP Protocol to delegate the user's credentials. For more details about how and when these protocols trigger the CredSSP Protocol, refer to [MS-RDPBCGR] or [MS-WSMV].

As described in [MS-CSSP], the CredSSP Protocol first establishes a Transport Layer Security (TLS)-encrypted channel between the client and the target server by using the SSL/TLS Protocol [MS-TLSP]. The CredSSP Protocol uses TLS as an encrypted pipe; it does not rely on the client or server authentication services that are available in TLS. The CredSSP Protocol then uses the SPNEGO Protocol [MS-SPNG] and [MS-NEGOEX] to negotiate the NTLM or Kerberos authentication protocol that performs mutual authentication and provides confidentiality services, which are used to securely bind to the TLS channel and encrypt the credentials for the target server. In environments where the Kerberos protocol is not supported, the NTLM protocol is selected to establish trust

between the client computer and the server computer. Otherwise, the Kerberos authentication protocol is selected because the Kerberos protocol ensures server authentication.

The Kerberos Key Distribution Center (KDC) Proxy Protocol [MS-KKDCP] is used to allow Kerberos clients to use KDC proxy servers to communicate to KDCs for Kerberos Network Authentication Service (V5) protocol [RFC4120] and Kerberos change password protocol exchanges [RFC3244]. RDP and IPsec are examples of KDC proxy usage.

When a Kerberos client does not have connectivity to the KDC, but the client could use Kerberos to authenticate to an application server, Kerberos normally fails. With a Kerberos proxy client on the client host and a KDC proxy server with connectivity to the KDC, Kerberos authentication can be used to authenticate to the application server. The Kerberos proxy client sends standard Kerberos Authentication Service, ticket-granting service (TGS), and change password requests in HTTPS messages to the KDC proxy server. The KDC proxy server locates the KDC, sends these messages to the located KDC, and returns the received replies via HTTPS to the client.

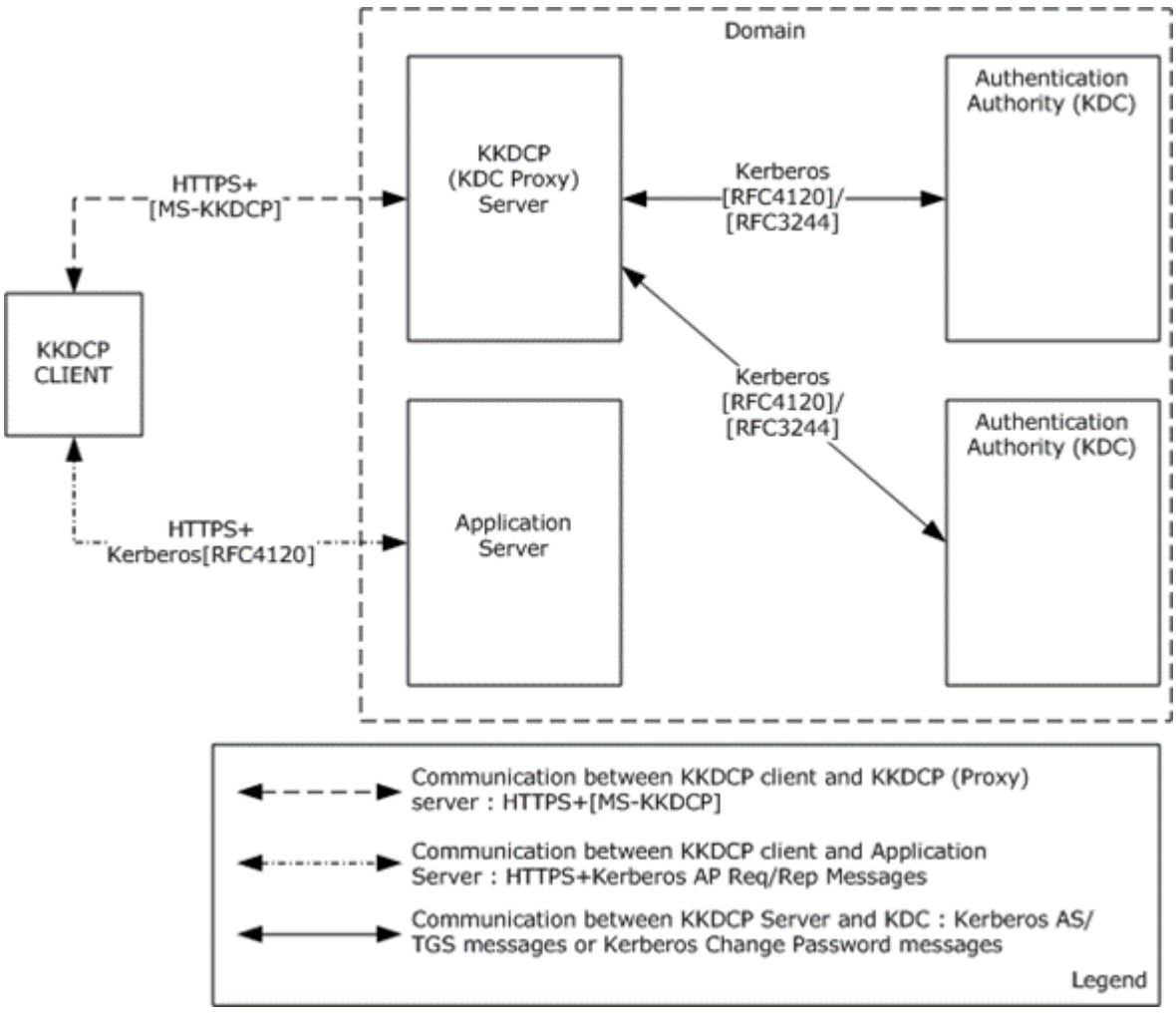


Figure 10: KKDCP deployment environment

Last updated on 10/30/2024